Classical Field Theory — Exercise Sheet 1

Local actions

Solutions

Use Lecture 1's sign $E_a = \partial L/\partial\phi^a - \partial_\mu p_a^\mu$. All fields and prescribed sources in this sheet are smooth. The first three problems use only differentiation, integration by parts and the fundamental lemma. The final problem also uses finite-dimensional linear ODEs.

Exercise 1.1 (A complex source and two real variations). On $\mathbb{R}^{1,1}$ let $\Phi = \phi_1 + \mathrm{i}\phi_2$ be a complex field, $J = J_1 + \mathrm{i}J_2$ a prescribed complex source, and $m \geq 0$. The density is

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\Phi_x|^2 - m^2|\Phi|^2\big) + \mathrm{Re}(\overline{J}\Phi).$$

For the action, take any compact region $D \subset \mathbb{R}^{1,1}$ and variations H whose support lies in the interior of D.

(a) Compute both real momentum currents and Euler expressions. Express the stationarity condition as one complex equation and derive the first variation for a compactly supported complex variation H.

(b) Find the source for which $\Phi(t,x) = (t^2+1) + \mathrm{i}(t^3 - x^2)$ is stationary.

(c) If only real-valued variations H are tested, must the complex equation hold? Give a counterexample with $m = 0$ and $J = 0$.

Solution. For the first part, the density in real components is

$$L = \frac{1}{2}\sum_{a=1}^{2}\big((\partial_t\phi_a)^2 - (\partial_x\phi_a)^2 - m^2\phi_a^2\big) + J_1\phi_1 + J_2\phi_2.$$

The definition of the momentum current and then the Euler expression give

$$p_a^0 = \partial_t\phi_a, \qquad p_a^1 = -\partial_x\phi_a, \qquad E_a = J_a - \big(\partial_t^2 - \partial_x^2 + m^2\big)\phi_a.$$

Let $E = E_1 + \mathrm{i}E_2$ and $H = H_1 + \mathrm{i}H_2$. The real first-variation formula is

$$\delta S_D[\Phi]H = \int_D (E_1H_1 + E_2H_2)\,\mathrm{d}t\mathrm{d}x = \mathrm{Re}\int_D \overline{E}H\,\mathrm{d}t\mathrm{d}x.$$

Tests with $H_2 = 0$ imply $E_1 = 0$, and tests with $H_1 = 0$ imply $E_2 = 0$. Together these are exactly the stated complex equation. No independent variation of two conjugate complex variables was assumed.

For the second part, apply the same differential operator to the actual field:

$$(\Box + m^2)(t^2+1) = 2 + m^2(t^2+1), \qquad (\Box + m^2)(t^3 - x^2) = 6t + 2 + m^2(t^3 - x^2).$$

These are the required two source components.

For the final part, the field $\Phi = \mathrm{i}t^2$ has $E = -2\mathrm{i}$. Its first variation against every real compact H vanishes because $\mathrm{Re}(\overline{E}H) = 0$. Its imaginary Euler expression is nonzero, so it is not stationary against all complex variations.

Exercise 1.2 (Natural conditions at the two spatial ends). Let $T, \ell > 0$, $m \geq 0$ and $\beta_0, \beta_\ell \geq 0$. On the rectangle $D = [0,T] \times [0,\ell]$, a real field has action

$$S_D[\phi] = \int_0^T \int_0^\ell \left(\frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - \frac{1}{2}m^2\phi^2 + J\phi \right) \mathrm{d}x\mathrm{d}t \\ - \frac{1}{2}\int_0^T \left(\beta_0 \phi(t,0)^2 + \beta_\ell \phi(t,\ell)^2\right) \mathrm{d}t,$$

where J is prescribed. Variations vanish near $t = 0, T$ and are unrestricted at $x = 0, \ell$.

(a) Integrate coordinate by coordinate and prove that stationarity is equivalent to the bulk equation and two Robin boundary conditions. Include the case $\beta_0 = \beta_\ell = 0$.

(b) For $\ell = 1$, $m = 0$, $J = 2$ and $\beta_0 = \beta_\ell = 1$, find the unique time-independent stationary field.

Solution. The variation of the bulk density is $\phi_t H_t - \phi_x H_x - m^2\phi H + JH$. Time integration has no endpoint term by the hypothesis on H. Spatial integration and the two boundary-action variations therefore give

$$\delta S_D[\phi]H = \int_0^T \int_0^\ell \left(J - \phi_{tt} + \phi_{xx} - m^2\phi\right) H \, \mathrm{d}x\mathrm{d}t \\ + \int_0^T \left(\phi_x(t,0) - \beta_0\phi(t,0)\right) H(t,0) \mathrm{d}t \\ + \int_0^T \left(-\phi_x(t,\ell) - \beta_\ell\phi(t,\ell)\right) H(t,\ell) \mathrm{d}t.$$

This derives the rectangle formula directly, rather than importing the lecture's torus-graph formula on a different region. Interior tests force the bulk coefficient to vanish. Given any compactly supported smooth time function g, the tests $H(t,x) = g(t)(1 - x/\ell)$ and $H(t,x) = g(t)x/\ell$ select the two traces independently. The one-dimensional fundamental lemma forces each boundary coefficient to vanish. Conversely, the bulk equation and both conditions make the displayed variation zero for every allowed H. If both coefficients β are zero, the conditions reduce to Neumann conditions $\phi_x = 0$ at both ends.

For the second part, write the static field as $u(x)$. The equation $-u'' = 2$ gives

$$u(x) = -x^2 + Ax + B.$$

The left condition gives $A = B$. The right condition gives $-2 + A = -(-1 + A + B)$, so $2A + B = 3$. Thus $A = B = 1$, uniquely. Both endpoint values are one and the derivatives are one and minus one, respectively, as the Robin conditions require.

Exercise 1.3 (Lorentz changes of coordinates). The Lorentz group in two coordinates is the matrix group

$$O(1,1) = \{\Lambda \in GL(2,\mathbb{R}) : \Lambda^{\mathsf{T}} \eta \Lambda = \eta\}, \qquad \eta = \operatorname{diag}(1,-1).$$

For $|v| < 1$, set

$$\gamma_v = (1-v^2)^{-1/2}, \qquad \begin{pmatrix} t' \\ x' \end{pmatrix} = \Lambda_v \begin{pmatrix} t \\ x \end{pmatrix}, \qquad \Lambda_v = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}.$$

(a) Verify that $\Lambda_v \in O(1,1)$ and $\det \Lambda_v = 1$.

(b) For a scalar field, set $\phi'(t',x') = \phi(t,x)$ at corresponding points. Use the chain rule to prove invariance of $\partial_t^2 - \partial_x^2$ and of $\frac{1}{2}(\phi_t^2 - \phi_x^2) - V(\phi)$, where V is any smooth function. Conclude equality of the scalar actions on corresponding compact regions.

Solution. Matrix multiplication gives

$$\Lambda_v^{\mathsf{T}} \eta \Lambda_v = \gamma_v^2 \begin{pmatrix} 1-v^2 & 0 \\ 0 & -(1-v^2) \end{pmatrix} = \eta, \qquad \det \Lambda_v = \gamma_v^2 (1-v^2) = 1.$$

The inverse is Λ_{-v}, so the chain rule gives

$$\partial_{t'} = \gamma_v(\partial_t + v\partial_x), \qquad \partial_{x'} = \gamma_v(v\partial_t + \partial_x).$$

Mixed derivatives commute for a smooth scalar. Expanding the operator difference gives

$$\partial_{t'}^2 - \partial_{x'}^2 = \gamma_v^2\big((1-v^2)\partial_t^2 - (1-v^2)\partial_x^2\big) = \partial_t^2 - \partial_x^2.$$

The same algebra applied to the first derivatives gives

$$(\phi'_{t'})^2 - (\phi'_{x'})^2 = \phi_t^2 - \phi_x^2.$$

At corresponding points $V(\phi') = V(\phi)$. Thus the densities agree. Since $\mathrm{d}t'\mathrm{d}x' = \mathrm{d}t\mathrm{d}x$ and $D' = \Lambda_v D$, change of variables yields

$$\int_{D'} L[\phi']\,\mathrm{d}t'\mathrm{d}x' = \int_D L[\phi]\,\mathrm{d}t\mathrm{d}x.$$

This statement compares the transformed field and region together.

Exercise 1.4 ⋆. Let $N \geq 3$, $h_{\mathrm{lat}} = 2\pi/N$, and let $q_j(t)$, $j = 0, \dots, N-1$, be real coordinates with periodic indexing. The chain Lagrangian is

$$L_h = h_{\mathrm{lat}} \sum_{j=0}^{N-1} \left[\frac{1}{2}\dot{q}_j^2 - \frac{1}{2h_{\mathrm{lat}}^2}(q_{j+1} - q_j)^2 - \frac{1}{2}m^2 q_j^2 \right], \qquad m \geq 0.$$

(a) Derive its equations by varying every q_j.

(b) Let $T > 0$ and let ϕ be a given smooth periodic solution of $\phi_{tt} - \phi_{xx} + m^2\phi = 0$ on $[0,T] \times \mathbb{T}$. Assume $M_4 \geq 0$ and $|\partial_x^4 \phi| \leq M_4$ there. Prove that sampling this field in the chain equation produces a residual bounded by $h_{\mathrm{lat}}^2 M_4/12$ at every node.

(c) Give the chain the exact sampled initial position and velocity. Set $e_j = q_j - \phi(t, jh_{\mathrm{lat}})$, $D_h e_j = (e_{j+1} - e_j)/h_{\mathrm{lat}}$ and $\|w\|_h^2 = h_{\mathrm{lat}} \sum_j w_j^2$. With $A_h = \sqrt{2\pi} M_4 h_{\mathrm{lat}}^2/12$, prove, for $0 \leq t \leq T$,

$$\left(\|\dot{e}(t)\|_h^2 + \|D_h e(t)\|_h^2 + m^2 \|e(t)\|_h^2 \right)^{1/2} \leq A_h t, \qquad \|e(t)\|_h \leq \frac{1}{2} A_h t^2.$$

The finite linear chain has a global solution by the linear ODE theorem. No existence theorem for the continuum equation is assumed beyond the given ϕ.

Solution. For the first part, variation of the two springs meeting at node j gives

$$\frac{\partial L_h}{\partial q_j} = \frac{q_{j+1} - 2q_j + q_{j-1}}{h_{\mathrm{lat}}} - h_{\mathrm{lat}} m^2 q_j, \qquad \frac{\partial L_h}{\partial \dot{q}_j} = h_{\mathrm{lat}} \dot{q}_j.$$

Subtracting the time derivative of the second expression from the first and dividing by $-h_{\mathrm{lat}}$ gives the chain equation.

For consistency, define $\Delta_h w_j = (w_{j+1} - 2w_j + w_{j-1})/h_{\mathrm{lat}}^2$. Taylor's theorem at a fixed time gives

$$\phi(x \pm h_{\mathrm{lat}}) = \phi(x) \pm h_{\mathrm{lat}} \phi_x(x) + \frac{h_{\mathrm{lat}}^2}{2}\phi_{xx}(x) \pm \frac{h_{\mathrm{lat}}^3}{6}\phi_{xxx}(x) + r_{\pm}, \qquad |r_{\pm}| \leq \frac{h_{\mathrm{lat}}^4 M_4}{24}.$$

The odd terms cancel on addition. Therefore

$$|\Delta_h \phi(t,x) - \phi_{xx}(t,x)| \leq \frac{h_{\mathrm{lat}}^2 M_4}{12}.$$

The sampled residual is $R_j = \phi_{tt}(t, jh_{\mathrm{lat}}) - \Delta_h \phi(t, jh_{\mathrm{lat}}) + m^2 \phi(t, jh_{\mathrm{lat}}) = \phi_{xx}(t, jh_{\mathrm{lat}}) - \Delta_h \phi(t, jh_{\mathrm{lat}})$. Since $Nh_{\mathrm{lat}} = 2\pi$, the bound gives $\|R(t)\|_h \leq A_h$.

For the error estimate, subtraction gives $\ddot{e} - \Delta_h e + m^2 e = -R$, with $e(0) = \dot{e}(0) = 0$. Periodic reindexing proves the summation identity

$$h_{\mathrm{lat}} \sum_j v_j \Delta_h e_j = -h_{\mathrm{lat}} \sum_j (D_h v_j)(D_h e_j).$$

Define the discrete error energy by

$$\mathcal{E}_h(t) = \frac{1}{2}\left(\|\dot{e}\|_h^2 + \|D_h e\|_h^2 + m^2 \|e\|_h^2 \right).$$

Its derivative, using the identity and the error equation, is

$$\mathcal{E}_h' = -h_{\mathrm{lat}} \sum_j \dot{e}_j R_j \leq \sqrt{2\mathcal{E}_h}\, \|R\|_h.$$

For $\epsilon > 0$, differentiation gives $\frac{\mathrm{d}}{\mathrm{d}t}\sqrt{2\mathcal{E}_h + \epsilon^2} \le \|R\|_h$. Integrate from zero and let $\epsilon \downarrow 0$ to obtain $\sqrt{2\mathcal{E}_h(t)} \le A_h t$. Finally, integrate $\dot{e}$ in the finite-dimensional norm:

$$\|e(t)\|_h \le \int_0^t \|\dot{e}(s)\|_h \mathrm{d}s \le \int_0^t A_h s \mathrm{d}s = \frac{1}{2} A_h t^2.$$

The last step supplies the full position bound even when the mass is zero. The estimate compares a given smooth continuum solution to its sampled chain and proves the stated finite-time convergence.